

Polynomials – 2

Integer coefficients, rational roots and the shape of the coefficients

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Theoretical minimum.

Theorem 1 (the two arithmetic facts). For $f \in \mathbb{Z}[x]$ and integers $a \neq b$ one has $a - b \mid f(a) - f(b)$. If p/q in lowest terms is a root of $c_n x^n + \dots + c_0 \in \mathbb{Z}[x]$, then $p \mid c_0$ and $q \mid c_n$. Together these two remarks settle most questions about integer and rational values of a polynomial.

Theorem 2 (Gauss and conjugates). A polynomial in $\mathbb{Z}[x]$ which factors over $\mathbb{Q}$ factors over $\mathbb{Z}$ with factors of the same degrees. If $f \in \mathbb{Q}[x]$ is irreducible and α is a root of f , then every symmetric rational expression in the roots of f is rational, and any rational relation satisfied by α is satisfied by all conjugates of α .

Fact 1 (coefficients). The coefficients of f are recovered from its values by finite differences, and sums such as $\sum a_k^2$ can be read off as $\frac{1}{2\pi} \int_0^{2\pi} |f(e^{it})|^2 dt$. This turns questions about coefficients into questions about values on the unit circle.

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1. (Putnam 1940) Prove that if $f(x)$ is a polynomial with integer coefficients and there is an integer k such that none of $f(1), f(2), \dots, f(k)$ is divisible by k , then f has no integer root.
 2. (Putnam 2004) Let $P(x) = \sum_{k=0}^n c_k x^k$ have integer coefficients and let r be a rational number with $P(r) = 0$. Show that the n numbers $c_n r, c_n r^2 + c_{n-1} r, \dots, c_n r^n + c_{n-1} r^{n-1} + \dots + c_1 r$ are integers.
 3. (Putnam 2000) Let $f(x)$ have integer coefficients and define integers $a_0 = 0$ and $a_{n+1} = f(a_n)$. Prove that if $a_m = 0$ for some $m \geq 1$, then $a_1 = 0$ or $a_2 = 0$.

4. (Putnam 2001) For each integer m consider $P_m(x) = x^4 - (2m+4)x^2 + (m-2)^2$. For what values of m is P_m a product of two non-constant polynomials with integer coefficients?
5. (Putnam 2003) Let $f(z) = az^4 + bz^3 + cz^2 + dz + e = a(z-r_1)(z-r_2)(z-r_3)(z-r_4)$ with integer a, b, c, d, e and $a \neq 0$. Show that if $r_1 + r_2$ is rational and $r_1 + r_2 \neq r_3 + r_4$, then $r_1 r_2$ is rational.